

Darsh Thakkar

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Professional Summary

Software engineer with experience building full-stack applications, AI/ML systems, backend services, and robotics interfaces. Skilled in Python, C++, React, APIs, databases, and production-oriented testing, with hands-on engineering experience across research, internships, and technical projects.

Experience

Undergraduate Research Assistant — People and Robots Lab, UW-Madison

Feb 2026 - Present | Madison, WI, USA

- Developed a Flask-based failure-handling pipeline that converts Temi robot failure events, including blocked obstacles, low battery, and location changes, into structured LLM-generated fallback behaviors.
- Built React components for a robot programming interface with real-time status visualization, user feedback collection, condition switching, and retry-state handling.
- Deployed and validated the system on a physical Temi robot, integrating the React interface, Flask backend, and Android application to execute and test robot failure scenarios end-to-end.

AI and Software Engineer Intern — LexLegis AI

May 2025 - Jul 2025 | Mumbai, India

- Developed a fully offline AI legal assistant with chat, OCR, multilingual translation, and semantic search to ensure data privacy.
- Built cross-platform desktop features in C++, Python, and Qt/QML; integrated local LLM inference via llama.cpp and Llama Server.
- Implemented hybrid retrieval pipelines combining BM25 and vector embeddings to ground locally generated responses across document collections.

Software Development Intern — Yodaplust

Jun 2024 - Aug 2024 | Mumbai, India

- Debugged SQL queries and backend behavior, identifying data and application issues and collaborating with engineers on fixes.
- Wrote Pandas-based data cleaning/validation scripts to improve data quality for downstream analytics and APIs.
- Developed full-stack features using Python, REST APIs, PostgreSQL, and React.

Projects

LunaYield — AI-Assisted Lunar Mission Planning & Safety Platform | IBM August AI Builders Challenge | [GitHub](#)

- Built and deployed a full-stack lunar mission planning platform that monitors rover resources, detects anomalies, and generates alternative recovery strategies while balancing scientific yield against safe mission completion.
- Designed a deterministic backend-authoritative safety engine that validates recovery plans against a 20% minimum return-battery threshold, rejects unsafe strategies, and requires operator approval before execution.
- Implemented real-time telemetry, WebSocket synchronization, audit logging, persistence, and per-session mission isolation using FastAPI, React, TypeScript, SQLAlchemy, and SQLite; validated with 539 automated tests.

ShadowOps — AI Deployment Preflight Platform | IBM July Wildcard Hackathon | [GitHub](#)

- Built and deployed a full-stack platform that reconstructs operational workflows from structured event logs, surfaces hidden human effort, and evaluates proposed AI automation before deployment.
- Designed a deterministic AI Tax model measuring five overhead categories, net savings, burden concentration, and deployment readiness; measured 20.5% overhead across a synthetic 46-event workflow.
- Integrated IBM Granite through watsonx.ai to generate grounded recommendations and safer hybrid workflows, adding PII redaction, structured output validation, cached fallback, and 60 backend tests.

Adaptive Rover-Drone Autonomy — Multi-Modal Robotics System | StarkHacks Purdue

- Built a cost-based autonomous decision engine that evaluates terrain, obstacle, and robot-state conditions to choose between continued rover navigation and a SWITCH_TO_DRONE transition.
- Integrated YOLOv8 perception with ROS 2 Humble on an NVIDIA Jetson Orin Nano, publishing object detections for downstream terrain evaluation and autonomous decision-making.
- Designed a modular robotics architecture supporting Manual, Semi-Autonomous, and Fully Autonomous modes, with extensible interfaces for LiDAR, Nav2 obstacle data, and IMU feedback.

Education

University of Wisconsin—Madison — Madison, WI | BS in Computer Science and Data Science | Expected May 2027

Dean's List: Fall 2023, Fall 2024

Skills

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, R, C

Frameworks & Backend: React, Django, Django REST Framework, FastAPI, Flask, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, SQLite

AI/ML: PyTorch, scikit-learn, OpenCV, MediaPipe, NLP, Reinforcement Learning

LLM/AI Systems: RAG, embeddings, BM25, llama.cpp, IBM Granite, watsonx.ai, structured outputs, Tesseract OCR

Developer Tools: Git, Docker, ROS 2, CMake, pytest, Qt/QML, FFmpeg

Certifications

Meta Back-End Developer Professional Certificate — Meta, 2026

Supervised Machine Learning — DeepLearning.AI / Stanford Online, 2025